Auditing Offline-to-Deployed Selection Transfer under Fixed Decision Interfaces

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Abstract

Offline decision-making pipelines often select models using validation scores computed on collected data, then deploy the selected model through a fixed decision interface such as threshold alerts, hysteresis rules, budgeted top-k actions, residual-warning screens, or replenishment rules. We investigate offline-to-deployed selection transfer: whether the model selected by offline validation remains the deployed-utility winner after the same fixed decision interface and deployment friction are applied. We introduce a report-card audit that records the offline-selected model, deployed-utility winner, agreement rate, deployed-suboptimal share, paired utility shortfall, and tie diagnostics. Across the studied tasks, offline-selected candidates frequently become deployed-suboptimal once switching or adjustment costs are applied through the fixed interface, and the mismatch recurs across seeds and interface checks. The report card serves as a pre-deployment selection audit for deciding when an offline winner should be reviewed under the intended interface and friction model before deployment, online evaluation, or adaptation.

1. Introduction

Offline decision-making pipelines often select a model using validation scores computed on collected data, then deploy the selected model through a fixed decision interface that maps predictions or scores into actions. Examples include triggering an alert when $p_t > \tau$, changing state only if the score moves by at least δ, issuing the top-k alerts under a fixed budget, applying a residual-warning screen, or applying a replenishment rule. When the deployed interface includes switching or adjustment friction, the model selected by the offline score need not remain the deployed-utility winner. We do not tune the interface per model: the same specified decision interface is applied to every candidate model before deployed-utility scoring.

This paper studies offline-to-deployed selection transfer as a pre-deployment selection audit. By selection transfer we mean whether the model selected by an offline validation score remains the best model under deployed utility after a fixed decision interface and friction model are applied. This is not an argument that offline validation metrics are invalid for their intended proxy objectives, nor a proposal for a single deployed metric. Deployed utility depends on a specified interface, simulator, and friction model. When offline validation is used as deployment-facing model-selection advice, the evaluation should also report whether that selection transfers to deployed utility under the specified fixed interface and friction model.

In offline decision-making terms, each candidate prediction model f_m, together with the fixed interface I_θ, induces a candidate decision rule $\pi_m = I_\theta \circ f_m$. The report card is therefore a fixed-candidate selection audit: it asks whether the candidate chosen by an offline proxy score is also preferred by deployed utility under the intended interface and friction model. In many offline decision-making settings, the first deployment decision is not how to update a policy online, but which offline-trained candidate is reliable enough to deploy, evaluate, or adapt. Our report card targets this pre-adaptation selection step.

This audit is complementary to OPE, decision-focused learning, and online adaptation: those methods estimate policy value, optimize downstream objectives, or adapt policies online, whereas our report card checks whether a fixed offline-selected candidate remains preferred under the intended interface and friction model.

This paper makes three focused contributions.

- Selection-transfer diagnostic. We formulate offline-to-deployed selection-transfer robustness: whether the model selected by an offline validation score remains the best model under deployed utility after a fixed decision interface and friction model are applied.
- Report-card artifact. We provide a lightweight report card recording offline-selected models, deployed-utility winners, agreement rates, deployed-suboptimal shares,

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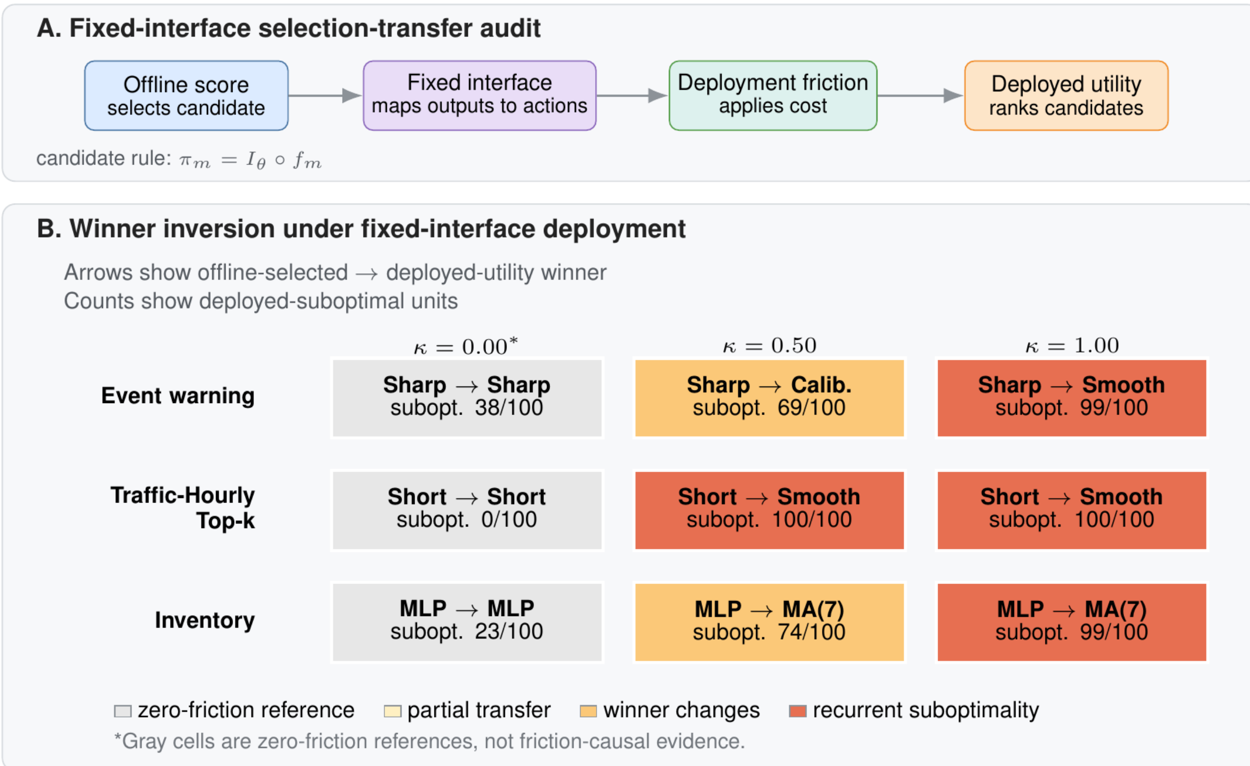

Figure 1. Offline-selected candidates can become deployed-suboptimal under fixed-interface deployment. (A) The audit selects candidates by an offline score, applies the same fixed decision interface to every candidate, and ranks resulting action paths by deployed utility under friction. (B) Arrows report aggregate offline-selected and deployed-utility winners; subopt. counts report seed- or unit-level cases where the offline-selected candidate is not deployed-best. Gray zero-friction cells are references, not friction-causal evidence.

paired utility shortfalls, and tie/uncertainty diagnostics.

- Empirical audit. Across prediction-to-decision tasks, we find recurrent selection failures in the tested higher-friction settings, while retaining zero-friction anchors, failed-gate variants, and tie audits to bound the interpretation.

Setup. For each candidate model $m \in \mathcal{M}$, an offline validation score $S_{\text{off}}(m)$ selects $m_{\text{off}} = \arg\min_m S_{\text{off}}(m)$. We use the loss convention for S_{off}, with lower values better. A fixed decision interface I_θ maps model outputs to actions, $a^m_{1:T} = I_\theta(\hat{y}^m_{1:T})$, and the same specified interface is applied to every candidate model. We evaluate deployed utility $U_\kappa(m; I_\theta)$ under friction κ, with larger values better. The deployed-utility winner is $m_{\text{dep}} = \arg\max_m U_\kappa(m; I_\theta)$. Selection transfer is preserved when $m_{\text{off}} = m_{\text{dep}}$; otherwise the offline-selected model incurs deployed shortfall $\Delta_\kappa = U_\kappa(m_{\text{dep}}; I_\theta) - U_\kappa(m_{\text{off}}; I_\theta)$. All deployed outcomes are reported as utilities or utility shortfalls, so larger U_κ is better and $\Delta_\kappa > 0$ means the offline-selected model is deployed-suboptimal. Near-ties are handled by the pre-specified tie-breaking rule and by the ϵ-tie audit reported in the appendix. Throughout the paper, "offline-selected model" denotes the candidate chosen by the offline validation score; "deployed-utility winner" denotes the candidate with the highest utility after the same fixed interface and friction model are applied.

The main audit question is whether, with the interface fixed, the offline-selected model remains the deployed-utility winner. A secondary mechanism check holds the proposed action path fixed and asks how frictional execution can change realized utility. We use this mechanism check only to explain why selection-transfer failures can arise.

2. Results

2.1. Selection transfer under fixed decision interfaces

This subsection answers the main audit question: with the decision interface fixed, does the offline-selected model remain the deployed-utility winner? Table 1 reports the selection-transfer report card. Event warning and Traffic-Hourly carry the primary selection-transfer evidence; EPA and inventory are bounded support.

Why offline ordering need not be preserved. Offline validation scores evaluate their specified validation objective, not the composed object obtained after model outputs are

Table 1. Offline-to-deployed selection-transfer report card. Each row fixes a task, interface, and friction level, then compares the offline-selected model with the deployed-utility winner. Transfer is agreement with the deployed winner; Subopt. counts seed- or evaluation-cell-level deployed-suboptimal cases. Shortfall is the mean paired deployed-utility gap in task-specific units.

Task	Fixed interface	κ	Offline-selected	Deployed winner	Transfer	Subopt.	Shortfall	95% CI	Evidence role
Synthetic	zero-friction anchor	0.00	Naive last	Naive last	1.00	0/20	0.000	[0.000, 0.000]	sanity
Event warning	threshold τ=0.55	0.50	Reactive sharp	Calibrated	0.31	69/100	0.011	[0.009, 0.014]	main audit
Event warning	threshold τ=0.55	1.00	Reactive sharp	Smoother	0.01	99/100	0.057	[0.052, 0.063]	main audit
Budgeted traffic alert	budget k=249	0.50	Reactive short	Smoother	0.00	100/100	7.02	[6.76, 7.28]	supporting audit
Budgeted traffic alert	budget k=249	1.00	Reactive short	Smoother	0.00	100/100	24.65	[24.23, 25.08]	supporting audit
PM2.5 warning	residual warning	1.00	Reactive lag-1	Long smoother	0.00	720/825	17.94	[16.56, 19.34]	residual support
Inventory replenishment	replenishment	1.00	Small MLP	MA(7)	0.01	99/100	1.03	[0.95, 1.11]	operational support

mapped to actions and executed through a frictional interface. If a candidate model's offline validation advantage is smaller than the extra friction-induced loss created by its implied action path, the deployed ordering can reverse even under a fixed interface. At the deployed-utility level, if candidate i has non-frictional advantage $\Delta_B > 0$ over j, but incurs additional frictional loss larger than Δ_B, then j can become the deployed-utility winner under the same fixed interface.

The event-warning microbenchmark is the primary controlled prediction-to-decision evidence because it isolates the selection-transfer question under a simple threshold interface and includes three robustness checks: an alternate threshold, log-loss reranking, and a hysteresis interface. In the report card, the offline-selected model remains Reactive sharp, but the deployed-utility winner changes to Calibrated baseline at friction 0.50 and Lagged smoother at friction 1.00. The deployed-utility consequence is recurrent: the offline-selected model is deployed-suboptimal in 69/100 and 99/100 seeds at these two friction levels. The result is not confined to the canonical threshold or Brier ranking: at frictions 0.50/1.00, deployed-suboptimality remains 77/100 and 98/100 under an alternate threshold, 74/100 and 99/100 under log-loss reranking, and 55/100 and 82/100 under a hysteresis interface, where the direction is preserved with smaller magnitude.

Traffic-Hourly Top-k is a prediction-to-decision task: forecasts are ranked by an offline validation score, but the fixed interface deploys only the top-k actions under a budget. The Traffic-Hourly pattern is not isolated to the canonical budget: appendix heatmaps show the same zero-friction alignment and positive-friction mismatch across $k \in \{125, 249, 500, 750\}$ and across canonical, earlier, and later temporal splits. In the selected Top-k setting ($k = 249$), Reactive short is both the offline-selected model and the deployed-utility winner at zero friction, with exact transfer 1.0. Once switching costs are introduced, however, the offline-selected model remains Reactive short while the deployed-utility winner shifts to Lagged smoother, and deployed-suboptimality reaches $100/100$ seeds at both friction 0.5 and friction 1.0.

EPA PM2.5 adds a broader residual-warning audit row without changing the evidence hierarchy. In expanded EPA AQS PM2.5 residual warning, Reactive lag-1 remains the offline-selected model, while Long smoother becomes the deployed-utility winner at $\kappa = 1.00$. We treat its units as report-card evaluation cells rather than independent iid samples. The main EPA row uses expanded report-card evaluation cells, while Appendix D.2 reports coarser residual-audit family units. We interpret the EPA row as a broader residual-warning audit rather than independent population-level evidence; it shows that the report-card failure mode is not unique to the Event or Traffic constructions. The appendix reports the full residual-candidate audit, including appendix-only residual-audit support and failed-gate context.

An ϵ-tie audit gives the same qualitative reading for the main prediction-to-decision rows: Traffic-Hourly Top-k and Event warning at $\kappa \in \{0.50, 1.00\}$, and expanded EPA at $\kappa = 1.00$, remain deployed-suboptimal under relative tie thresholds up to $\epsilon_{\mathrm{rel}} = 0.005$; inventory remains corroborative and lower-margin secondary checks are reported only in the appendix.

Intervals are descriptive summaries for report-card cells, not hypothesis tests for a broader population-level claim. Appendix ϵ-tie and mixed/failed-gate variants help guard against reading the main table as selected successes only.

Inventory is retained as operational corroboration: it shows that the same audit format applies to a replenishment interface, but it is not used as primary evidence for the offline decision-making claim. At friction 0.50 and 1.00, the offline-selected Small MLP is deployed-suboptimal in 74/100 and 99/100 seeds, respectively, under the fixed replenishment interface; the report card includes only the high-friction inventory row to keep this role bounded.

Mechanism support under fixed action paths. The mechanism check is not the main empirical claim. It uses synthetic and inventory fixed-action replay: when the proposed action path is held fixed, frictional execution can separate proposed and realized actions and change realized utility. The fixed-interface selection-transfer rows in Table 1 provide the paper's actual selection-transfer evidence.

Appendix robustness. Appendix checks are organized to answer whether the main rows are isolated successes: Event robustness varies threshold, scoring rule, and interface; Traffic robustness varies k and deployment window; mixed/failed-gate and tie-sensitivity rows are retained to show where the audit does not support a paper-facing claim. Appendix C adds a replay diagnostic over a fixed candidate set, not an online learning claim: in Traffic Top-k, small calibration segments substantially reduce the observed misselection under fixed-interface replay on held-out outcomes. Stronger-pool and residual-audit checks remain appendix-only, while additional Traffic time-of-day and relative-rank checks, inventory variants, and load-following variants are summarized as secondary context.

3. Discussion, Related Work, and Limitations

Offline policy evaluation and offline-to-deployment selection. OPE and policy-comparison methods estimate the value of a target policy from logged behavior data under coverage, propensity, or model assumptions (Dudik et al., 2011; Swaminathan and Joachims, 2015; Thomas and Brunskill, 2016). Offline RL instead learns or constrains policies from logged data before online interaction (Fujimoto et al., 2019; Kumar et al., 2020). Offline decision-making benchmarks such as D4RL and Design-Bench emphasize evaluation under offline-data constraints; our audit is complementary, checking selection transfer for fixed candidates and fixed deployed interfaces rather than proposing a new optimizer or benchmark suite (Fu et al., 2020; Trabucco et al., 2022). Our object is different: given a fixed candidate set, offline validation rule, decision interface, and friction model, the report card audits whether the candidate selected by the offline proxy remains the deployed-utility winner. Thus the report card targets a preceding selection-transfer step rather than introducing a new value estimator, policy-learning algorithm, or online adaptation method.

Decision-focused learning and predict-then-optimize. Decision-focused learning and predict-then-optimize change the training or optimization objective to improve downstream decisions (Donti et al., 2017; Elmachtoub and Grigas, 2022; Mandi et al., 2024). Our candidates are already trained and the interface is held fixed; the contribution is a reporting layer that checks whether the existing offline selection rule would choose a deployment-suboptimal candidate.

Prediction benchmarks and decision-facing selection. Many offline evaluation settings rank candidate models by predictive accuracy, calibration, or benchmark score before those predictions are passed into downstream decision rules. Forecasting and time-series benchmarks are one important example, comparing model families and evaluation protocols across live forecasting tasks and broad archives (Godahewa et al., 2021; Aksu et al., 2024; Karger et al., 2025). Proper scoring rules remain the right object when the target is predictive quality (Murphy, 1993; Gneiting and Raftery, 2007). Recent decision-facing forecast evaluation also shows that model rankings can change under downstream decision tasks (Raeth and Ludwig, 2025). Our question is complementary: when such offline rankings are used as deployment-facing model-selection advice under a fixed decision interface, should the evaluation also report whether the offline-selected candidate remains the deployed-utility winner?

Benchmark uncertainty and report-card intervals. Our uncertainty reporting follows the spirit of benchmark-variance and leaderboard-interpretation work (Bouthillier et al., 2021; Longjohn et al., 2025), but reports selection transfer and deployed shortfalls rather than aggregate leaderboard uncertainty alone. Deployment frictions such as switching costs, adjustment costs, and operational churn can change which model is preferable after actions are executed; the report card makes this effect visible at the model-selection level.

Scope and limitations. Our claim is evaluative: the report card is a pre-deployment selection audit, not a new training objective, universal utility function, or online adaptation algorithm. We do not claim that offline validation metrics are invalid for their intended predictive or proxy objectives, and low, moderate, and high friction are grid-relative labels rather than universal calibrations. Event warning and Traffic-Hourly provide the core fixed-interface selection-transfer evidence; EPA and inventory show residual and operational extensions without changing the evidence hierarchy, while inventory and load-following variants remain secondary context. These settings form a focused audit, not a comprehensive offline decision-making benchmark suite. Switching and adjustment costs are illustrative fixed frictions. Practically, low transfer, high deployed-suboptimal share, or a large paired shortfall should trigger friction-aware review before the offline winner is treated as deployment-facing advice.

Impact Statement

This paper presents work whose goal is to improve evaluation and deployment readiness for offline decision-making systems. It does not introduce a new deployed policy for a particular high-stakes domain; potential societal effects arise from how practitioners use audits to review model-selection decisions before deployment.

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Appendix: Audit Protocol and Robustness Checks

Report-card fields for offline-to-deployed selection transfer.

For each task family, the report-card artifact records:

(i) task definition, evaluation split, seed/split manifest, and candidate families;

(ii) offline validation score, tie-breaking rule, fixed decision interface, and interface parameters;

(iii) friction grid, including zero-friction or reference cells when applicable;

(iv) offline-selected model, deployed-utility winner, transfer rate, deployed-suboptimal share/count, and paired shortfall;

(v) uncertainty summaries, ϵ-tie sensitivity, and mixed or failed-gate variants when applicable.

A minimal reversal condition. Consider two candidates i and j evaluated under the same fixed interface. Let B_m denote the non-frictional deployed benefit of candidate m, and let $C_m(\kappa)$ denote its deployment friction cost, so that $U_\kappa(m) = B_m - C_m(\kappa)$. Suppose i has non-frictional deployed advantage $\Delta_B = B_i - B_j > 0$. The deployed winner reverses to j whenever

$$C_i(\kappa) - C_j(\kappa) > \Delta_B.$$

Thus a fixed interface can preserve the preferred candidate before friction while reversing the deployed ordering once the additional frictional cost induced by its action path exceeds its non-frictional advantage. This condition is a minimal accounting identity, not a generalization theorem.

Table A1. Review-facing robustness map. Each row links a common interpretation concern to the appendix diagnostic that bounds it.

Concern	Appendix evidence	Concern	Appendix evidence
threshold sensitivity	Event alternate τ	top-k sensitivity	Traffic k-grid heatmap
offline metric sensitivity	Event log-loss reranking	temporal robustness	Traffic canonical/earlier/later splits
interface sensitivity	Event hysteresis	cherry-picking concern	Mixed/failed-gate variants
near-tie concern	ϵ-tie audit	residual-warning generality	EPA/Capital/Wikimedia plus failed NYC/M5

For compactness, some appendix tables use shortened column labels; the surrounding captions define their paper-facing meaning.

Audit-status labels. Audit-status labels are descriptive diagnostics, not hypothesis-test outcomes. In selection-transfer tables, *Pass* means that the row meets the stated audit checks: reference alignment where applicable, recurrent positive-friction deployed-suboptimality, positive paired shortfalls with descriptive intervals, and stability under the ϵ-tie audit when applicable. *Boundary* marks directionally consistent but lower-margin or partially recurrent behavior; *Fail* marks failed reference alignment, lack of recurrence, or behavior dominated by objective mismatch. Diagnostic-specific labels are local to their captions.

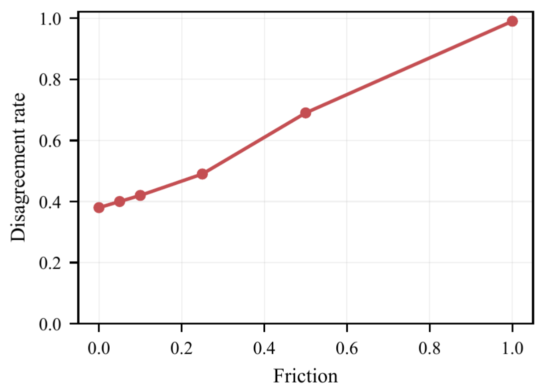
Mixed and failed-gate variants. Mixed and failed-gate variants are retained as audit context rather than main evidence, documenting where reference alignment, recurrence, or tie stability is insufficient for a paper-facing selection-transfer claim.

Table A2. Mixed, boundary, and failed-gate variants retained as audit context. These rows remain audit context rather than main report-card rows. They document where the audit is lower-margin, tie-sensitive, partially recurrent, or dominated by objective mismatch rather than friction-induced selection-transfer failure.

Variant	Audit status	Interpretation	Placement
Traffic daytime $\kappa = 0.5$	Boundary	Lower-margin, directionally consistent	Appendix only
Inventory zero/low friction	Mixed	Not a main zero-friction anchor	Appendix only
Expanded inventory	Mixed	Fairness context only	Appendix only
Load-alert screening	Boundary/Fail	Daily peak-hour Boundary; residual/four-week Fail	Screening artifacts
Residual-candidate audit	Mixed	EPA/Capital/Wikimedia satisfy residual-audit checks; NYC/M5 fail zero-friction alignment	Appendix D.2

A. Event-Warning Decision Task Robustness

Because the event-warning task provides the paper's main prediction-to-decision selection-transfer evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal binary-event setting evaluated over 100 seeds, in which each candidate prediction model emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical offline validation score. This appendix records the canonical six-row Brier summary plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.



κ	Offline-selected	Deployed winner	Transfer	Shortfall
0.00	Sharp	Sharp	0.62	0.003
0.05	Sharp	Sharp	0.60	0.002
0.10	Sharp	Sharp	0.58	0.002
0.25	Sharp	Calibrated	0.51	0.003
0.50	Sharp	Calibrated	0.31	0.011
1.00	Sharp	Smoother	0.01	0.057

Figure A1. Event warning canonical summary. Left: disagreement between offline and deployed-utility model selection increases with friction. Right: canonical fixed-threshold selection summary; shortfall is the mean paired deployed-utility shortfall of the offline-selected model.

Table A3. Alternate-threshold robustness for the event-warning task. The same qualitative selection-transfer mismatch and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Threshold	Friction	Offline-selected	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A4. Log-loss robustness for the event-warning task at the canonical threshold. The same qualitative selection-transfer mismatch and recurrent deployed-suboptimality pattern persists when offline validation ranking is recomputed with log loss instead of Brier.

Friction	Offline-selected	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A5. Second-interface robustness for the event-warning task. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative selection-transfer mismatch and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Interface	Friction	Offline-selected	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Across this appendix robustness package, the same qualitative selection-transfer mismatch and recurrent deployed-suboptimality pattern persists in a minimal prediction-to-decision event-warning task evaluated over 100 seeds.

A.1. Interface-Aware Event-Warning Stats Addendum

This addendum records interface-specific recurrence and gap summaries for the hysteresis-threshold check without changing the main cross-domain recurrence package.

The aggregate recurrence table reports the seed-level pattern directly, replacing the dense seed-strip visualization.

Table A6. Interface-aware recurrence and share uncertainty for event-warning deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Fixed threshold	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Fixed threshold	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Hysteresis threshold	0.50	100	0.45	0.55 (55/100)	[0.452, 0.644]	Wilson
Hysteresis threshold	1.00	100	0.18	0.82 (82/100)	[0.733, 0.883]	Wilson

Table A7. Interface-aware bootstrap intervals for the event-warning deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

B. Traffic Fixed-Budget Robustness

B.1. Interface/Friction Heatmap

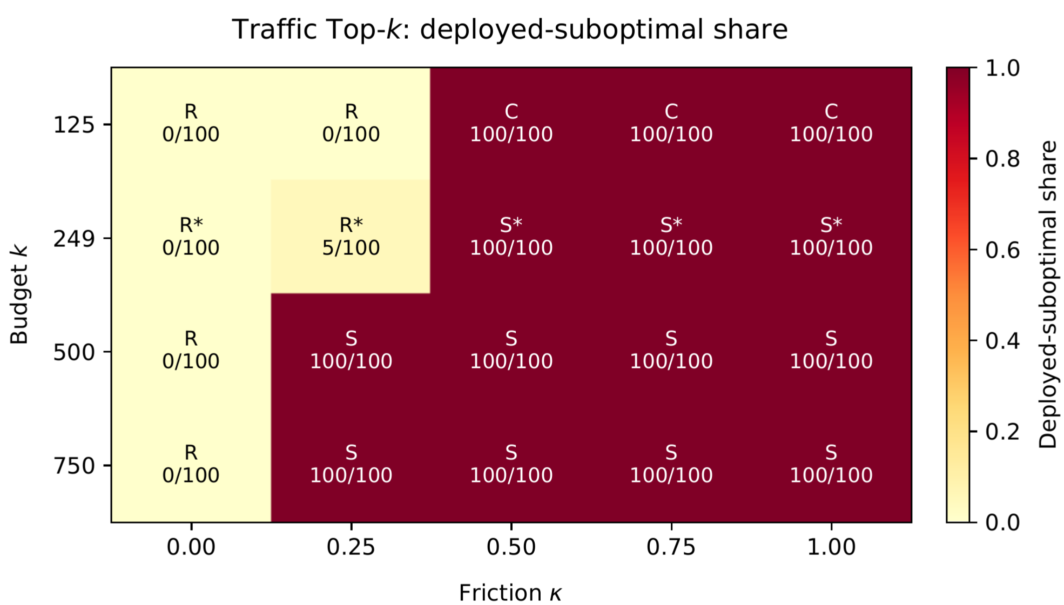


Figure A2. Traffic Top-k deployed-suboptimality across budgets and friction levels. Cell text reports the deployed winner label and deployed-suboptimal count of the offline-selected model: R = Reactive, S = Smoother, C = Calibrated; $*$ marks the canonical $k = 249$ budget.

The heatmap replaces the older compact Top-k family table: it extends the grid to $k = 750$ and intermediate frictions $\kappa \in \{0.25, 0.75\}$, while showing that the canonical $k = 249$ point is not isolated.

B.2. Temporal-Split Robustness

Traffic-Hourly temporal splits preserve the zero-friction alignment gate and show recurrent positive-friction deployed-suboptimality across canonical, earlier, and later deployment windows. Additional predefined Traffic time-of-day and relative-rank checks are summarized only as secondary audit context.

Table A8. Traffic-Hourly temporal-split robustness. Each split recomputes the offline-selected model from that split's validation segment. Zero-friction rows are alignment checks; positive-friction rows satisfy the audit checks when the split preserves zero-friction alignment and shows recurrent deployed-suboptimality with positive paired shortfalls. Because this temporal-split protocol recomputes validation and deployment windows within each split, its shortfalls need not numerically match the main canonical report-card row.

Split	κ	Offline sel.	Deploy winner	Subopt.	Mean shortfall	95% CI	Audit status
Canonical	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	Zero-friction aligned
Canonical	0.50	Reactive short	Lagged smoother	100/100	10.177	[9.871, 10.486]	Pass
Canonical	1.00	Reactive short	Lagged smoother	100/100	31.300	[30.823, 31.792]	Pass
Earlier	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	Zero-friction aligned
Earlier	0.50	Reactive short	Lagged smoother	100/100	9.152	[8.821, 9.487]	Pass
Earlier	1.00	Reactive short	Lagged smoother	100/100	27.861	[27.312, 28.423]	Pass
Later	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	Zero-friction aligned
Later	0.50	Reactive short	Lagged smoother	100/100	11.963	[11.654, 12.270]	Pass
Later	1.00	Reactive short	Lagged smoother	100/100	35.243	[34.764, 35.721]	Pass

C. Replay-Based Deployed-Feedback Reselection

This diagnostic asks whether a small calibration segment of deployed-feedback replay can reduce or correct an offline model-selection error within the same fixed candidate set and interface. It assumes replay access to held-out outcomes for all fixed-interface candidates and is used only as an appendix diagnostic. Calibration is always the earlier segment and evaluation is always the later segment; no future deployed utility from the evaluation segment is used when selecting the reselected candidate.

For compactness, Table A9 reports representative budget rows, including the lower-friction Event boundary case and the higher-friction Event and Traffic rows where replay-based reselection substantially reduces the observed misselection; the full budget grid remains in the isolated experiment report.

Table A9. Replay-based deployed-feedback reselection diagnostic. This diagnostic assumes replay access to held-out outcomes for all fixed-interface candidates. The lower-friction Event row is retained as boundary evidence: it improves shortfall but has nontrivial worse-than-offline share.

Domain	κ	Budget	Reduction	Resel. subopt.	Correction	Worse	Diagnostic status
Event	0.50	20%	28.7%	59%	41%	26%	Boundary
Event	1.00	10%	78.8%	30%	70%	2%	Pass
Traffic	0.50	10%	100%	0%	100%	0%	Pass
Traffic	1.00	10%	100%	0%	100%	0%	Pass

D. Candidate-Pool and Residual-Audit Checks

D.1. Stronger Candidate-Pool Fairness Pilot

This pilot checks whether the selection-transfer mismatch disappears under lightweight added candidates. It is not a forecasting model benchmark and uses fixed, lightweight configurations without extensive hyperparameter search.

Table A10. Expanded candidate-pool pilot. The qualitative selection-transfer mismatch remains under lightweight added candidates, but this check is appendix-only because the paper does not aim to benchmark forecasting model classes.

Domain	κ	Added candidates	Offline sel.	Deploy winner	Subopt.	Mean shortfall
Event	0.50	Logit/cal. logit/GBDT	Reactive sharp	Calibrated baseline	82/100	0.01884
Event	1.00	Logit/cal. logit/GBDT	Reactive sharp	Lagged smoother	97/100	0.06035
Traffic	0.50	Linear AR head	Reactive short	Lagged smoother	100/100	10.17694
Traffic	1.00	Linear AR head	Reactive short	Lagged smoother	100/100	31.29953

D.2. Residual-Warning Audit Family

We use this family as audit context rather than main-domain breadth: EPA, Capital Bikeshare, and Wikimedia satisfy the residual-audit alignment and recurrence checks, while NYC TLC and M5 fail the zero-friction alignment gate.

Table A11. Residual-warning audit family. EPA, Capital Bikeshare, and Wikimedia satisfy the residual-audit alignment and recurrence checks and are retained as appendix-only support; NYC TLC and M5 fail the zero-friction alignment gate and are retained as failed-gate context. The main EPA row reports expanded report-card evaluation cells; this table summarizes residual-candidate audit units at the coarser family level. These rows are not promoted as main-domain breadth.

Candidate	Audit status	Units	Zero subopt.	Positive subopt.	Mean shortfall	Placement
EPA AQS PM2.5	Pass	16	0.000	1.000	486.94	Appendix summary
Capital Bikeshare	Pass	16	0.063	1.000	2791.75	Appendix summary
Wikimedia hourly	Pass	11	0.000	1.000	118.18	Appendix summary
NYC TLC taxi-zone	Fail	16	0.750	1.000	2285.81	Failed-gate context
M5 store-quota	Fail	20	0.500	0.900	24.15	Failed-gate context

E. Mechanism, Tie Audit, and Secondary Corroboration

Inventory remains operational corroboration rather than a main-domain breadth claim: the main text reports the high-friction replenishment row, while mixed zero/low-friction and expanded-family inventory checks are retained only as secondary context. Load-following is likewise reduced to failed-gate context because its lower-margin rows are tie-sensitive or partially sensitive. Additional predefined Traffic time-of-day and relative-rank checks are summarized only as secondary audit context.

E.1. Mechanism Check under Fixed Action Paths

The mechanism check explains why selection-transfer divergence can arise. Holding the proposed action path fixed, synthetic and inventory show how a fixed interface can separate proposed and realized actions once friction is introduced.

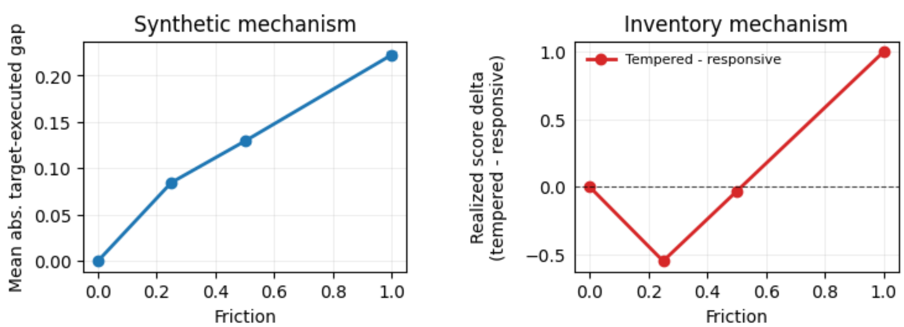


Figure A3. Mechanism support under fixed action paths. Even with the same proposed actions, a fixed frictional interface can alter realized actions and realized score. This illustrates how a fixed deployment interface can change the realized utility ordering of offline-selected candidates.

E.2. Uncertainty and ϵ-Tie Audit

This appendix reports uncertainty for the seed-level recurrence and paired deployed-shortfall summaries used in the main report card. We emphasize intervals and paired shortfalls rather than hypothesis-test significance. For continuous deployed shortfalls, we use 10,000 paired seed bootstrap resamples. For binary share summaries, rows with fewer than 30 matched seeds use Clopper–Pearson exact intervals; rows with at least 30 matched seeds use Wilson intervals. We report these intervals as descriptive uncertainty summaries for the report-card audit, not as hypothesis-test evidence for a broader population-level claim.

Table A12. Seed-level recurrence and share uncertainty for deployed-suboptimality. Agreement rate is the fraction of matched seeds for which the offline-selected model is also the deployed-utility best model; deployed-suboptimal share is the complementary fraction. We report Clopper–Pearson exact intervals for rows with $n < 30$ and Wilson intervals otherwise.

Domain	Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Event warning	threshold $\tau = 0.55$	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Event warning	threshold $\tau = 0.55$	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Traffic Top-k	budget $k = 249$	0.50	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Traffic Top-k	budget $k = 249$	1.00	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Inventory	replenishment	0.50	100	0.26	0.74 (74/100)	[0.646, 0.816]	Wilson
Inventory	replenishment	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson

Aggregate recurrence intervals and uncertainty bands summarize the seed-level pattern without the dense seed-strip visualization.

Table A13. Paired-bootstrap intervals for deployed-shortfall rows. Deployed shortfall is the paired deployed-utility shortfall $U_{\text{best}} - U_{\text{offline-selected}}$ computed within each matched seed. Intervals use 10,000 paired seed bootstrap resamples. Event warning and Traffic-Hourly provide the main prediction-to-decision selection-transfer evidence; inventory is operational corroboration.

Domain	Friction	Mean shortfall	Mean shortfall 95% bootstrap CI	Median shortfall	Median shortfall 95% bootstrap CI
Event warning	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event warning	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Traffic-Hourly Top-k	0.50	7.016	[6.756, 7.283]	6.977	[6.706, 7.281]
Traffic-Hourly Top-k	1.00	24.651	[24.226, 25.079]	24.473	[24.001, 25.051]
Inventory	0.50	0.240	[0.195, 0.286]	0.202	[0.154, 0.256]
Inventory	1.00	1.029	[0.946, 1.111]	1.045	[0.926, 1.167]

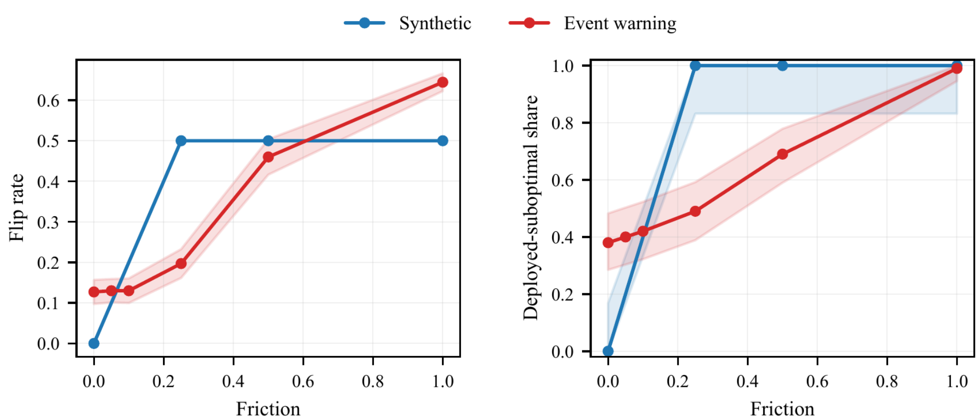


Figure A4. Selection-transfer uncertainty summaries for synthetic and event warning. Left: flip rate; right: deployed-suboptimal share. Bands follow the interval convention described in Appendix E.2; zero-width bands may coincide with the line.

E.3. Epsilon-Tie Sensitivity

The audit covers the gated report-card domains and is used to check whether deployed-suboptimality survives small relative tie thresholds.

Table A14. ϵ-tie sensitivity for the gated report-card domains. A seed is tie-compatible when the deployed shortfall of the offline-selected model is at most $\epsilon_{\text{rel}} S + 10^{-12}$, where S is the median seed-level deployed-utility scale within the row. Strict is the number of seeds still deployed-suboptimal after the epsilon tie threshold. The Tie status column records whether the deployed-suboptimality conclusion remains after applying the stated relative tie threshold; it is a descriptive robustness label, not a hypothesis-test outcome.

Domain	Interface	κ	ϵ_{rel}	n	S	Nom. subopt.	Adj. subopt.	Tie-comp.	Strict	Median Δ	Mean Δ	Tie status
Event warning	threshold $\tau = 0.55$	0.50	0.000	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event warning	threshold $\tau = 0.55$	0.50	0.001	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event warning	threshold $\tau = 0.55$	0.50	0.005	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event warning	threshold $\tau = 0.55$	1.00	0.000	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Event warning	threshold $\tau = 0.55$	1.00	0.001	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Event warning	threshold $\tau = 0.55$	1.00	0.005	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Traffic Top-k	k=249	0.50	0.000	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top-k	k=249	0.50	0.001	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top-k	k=249	0.50	0.005	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top-k	k=249	1.00	0.000	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Traffic Top-k	k=249	1.00	0.001	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Traffic Top-k	k=249	1.00	0.005	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Inventory	replenishment	0.50	0.000	100	4.841	0.74	0.74	26	74	0.202	0.240	stable
Inventory	replenishment	0.50	0.001	100	4.841	0.74	0.73	27	73	0.202	0.240	stable
Inventory	replenishment	0.50	0.005	100	4.841	0.74	0.72	28	72	0.202	0.240	stable
Inventory	replenishment	1.00	0.000	100	7.141	0.99	0.99	1	99	1.045	1.029	stable
Inventory	replenishment	1.00	0.001	100	7.141	0.99	0.99	1	99	1.045	1.029	stable
Inventory	replenishment	1.00	0.005	100	7.141	0.99	0.99	1	99	1.045	1.029	stable